

Parallelizing QEM Mesh Simplification

Team 24

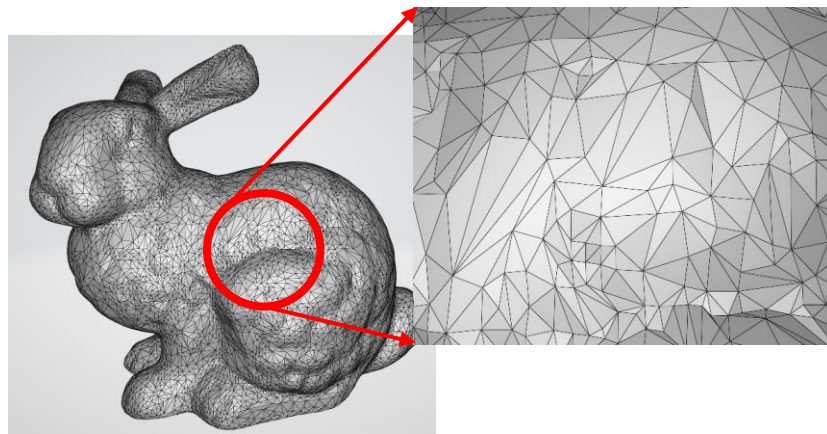
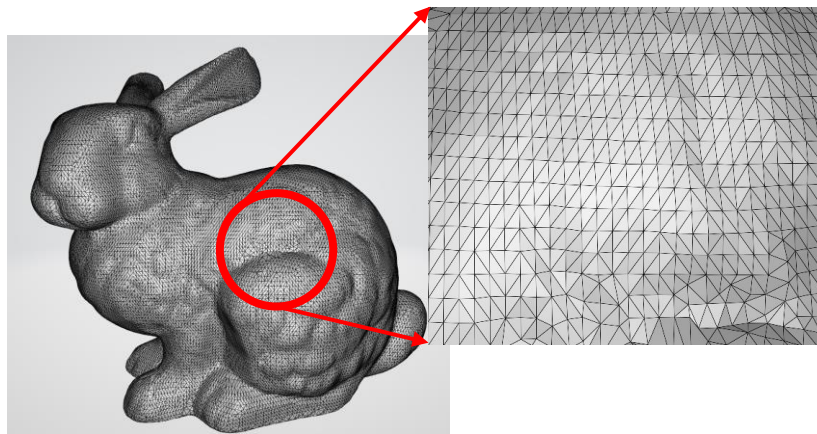
313551124 陳禹辰 313551138 劉昀翰 313553003 張綺恩

Outline

- Motivation and Objectives
- Methodology
- Experiments and Analysis
- Conclusion

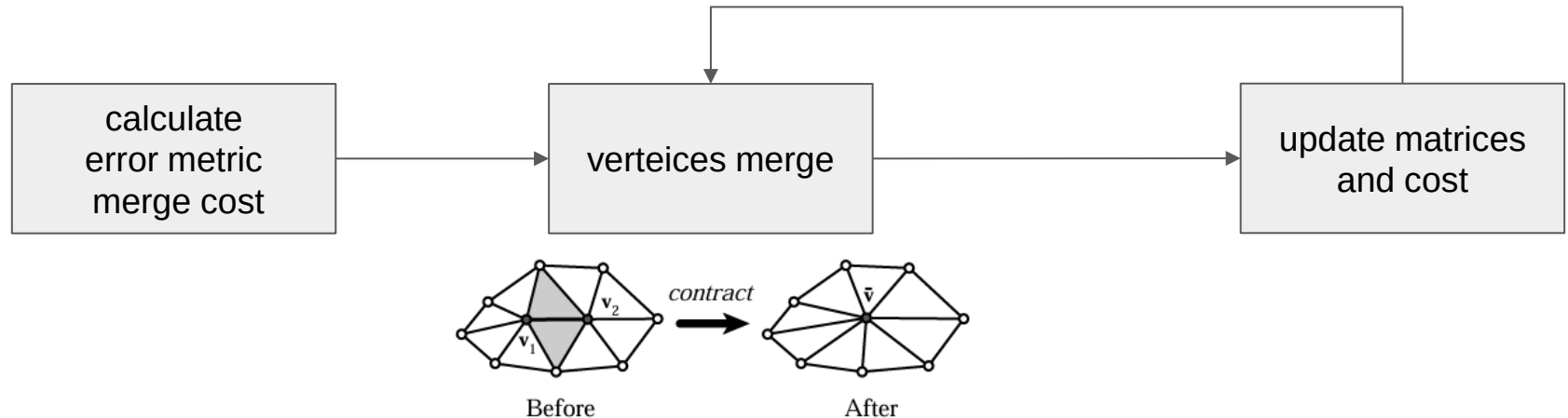
Motivation and Objectives

- Mesh Simplification
 - Reduces polygon count while preserving geometric features
 - Enhances rendering efficiency, widely used in games and VR



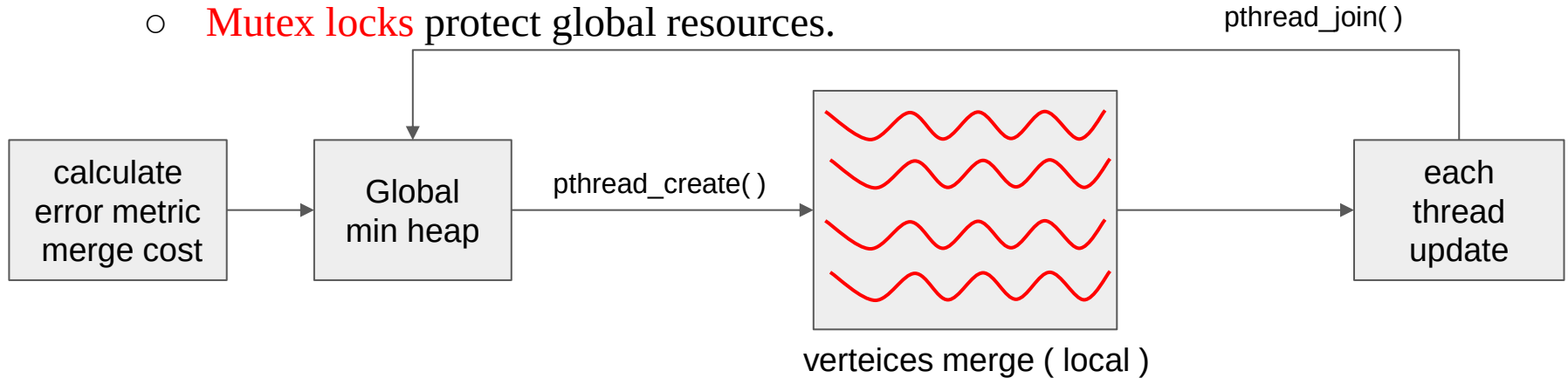
Methodology - Serial QEM

- Calculate a Q error matrix for each vertex and merge cost for each edge
- Merge edges with the smallest merge cost
- Update Q error matrices and merge cost after each merge
- Repeat until reaching the target simplification level



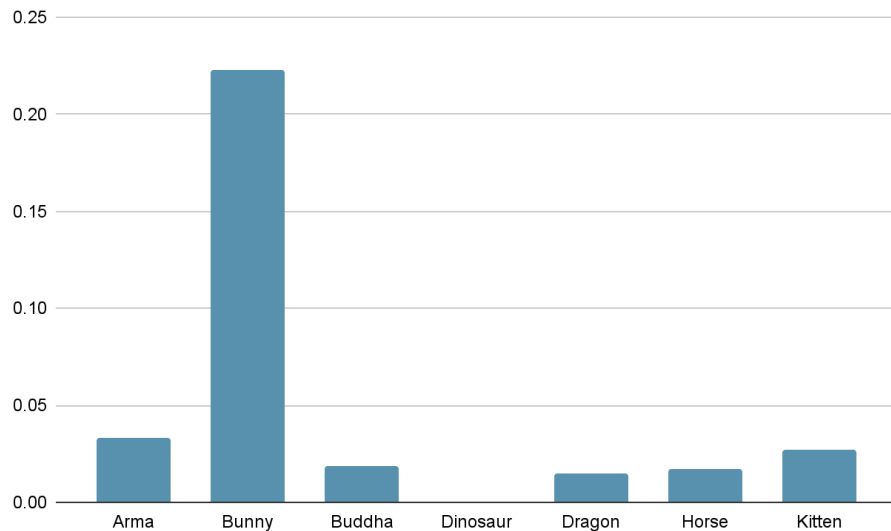
Methodology - Pthread

- Edge priority queue
 - Threads compute edge costs and store them in **local** queues.
 - **Local** queues are merged into a **global** queue.
- Edge Contraction
 - Threads merge vertices and update costs from local queues.
 - **Mutex locks** protect global resources.



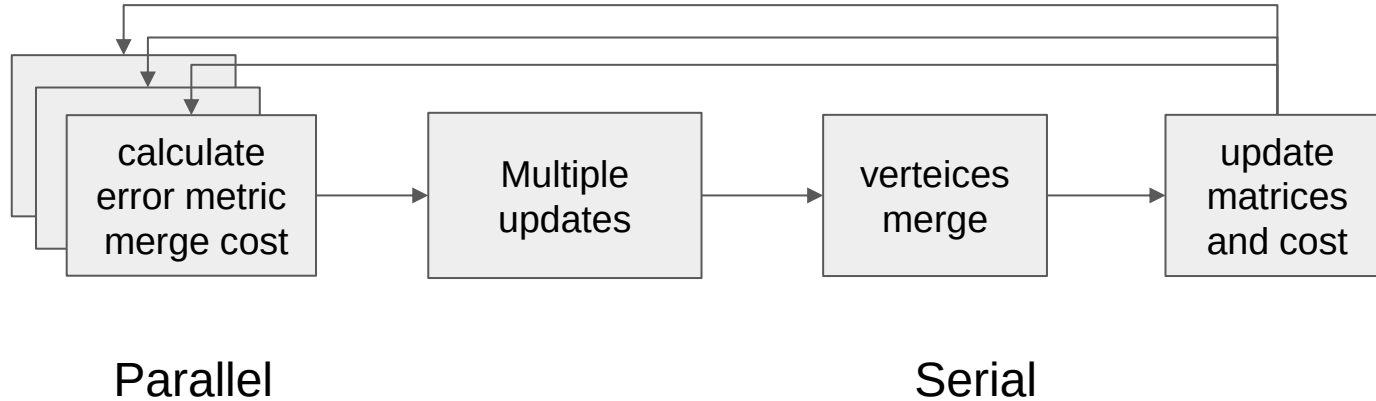
Methodology - Cuda

- Simple move the calculation of Q Matrix and position update to GPU
- Structure change to accommodate for the relocation of computation



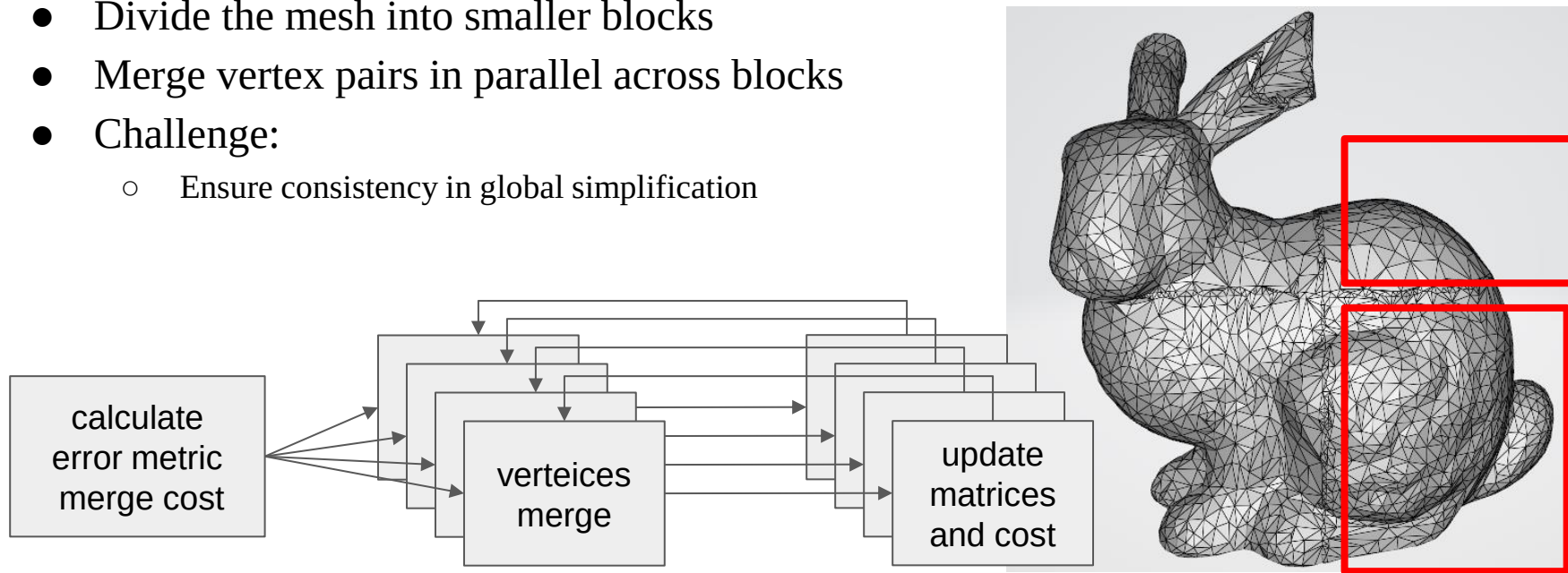
Methodology - Cuda

- Update multiple vertices in one iteration.
- Parallel computation of the updated edge's vertices

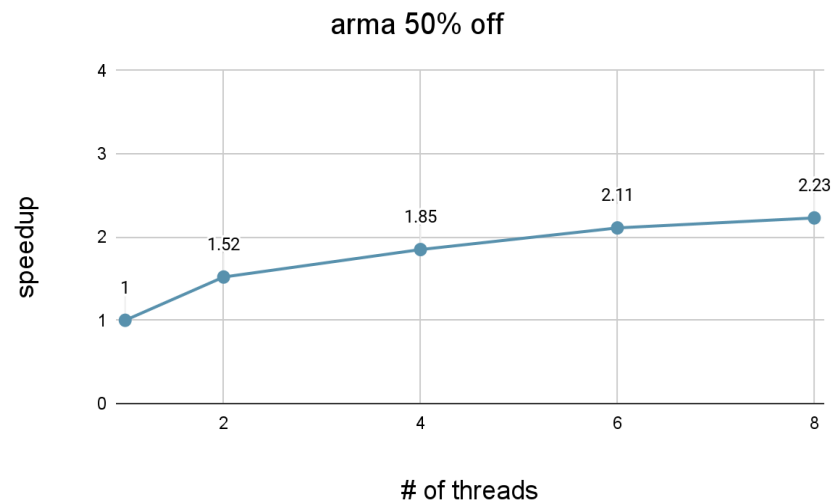
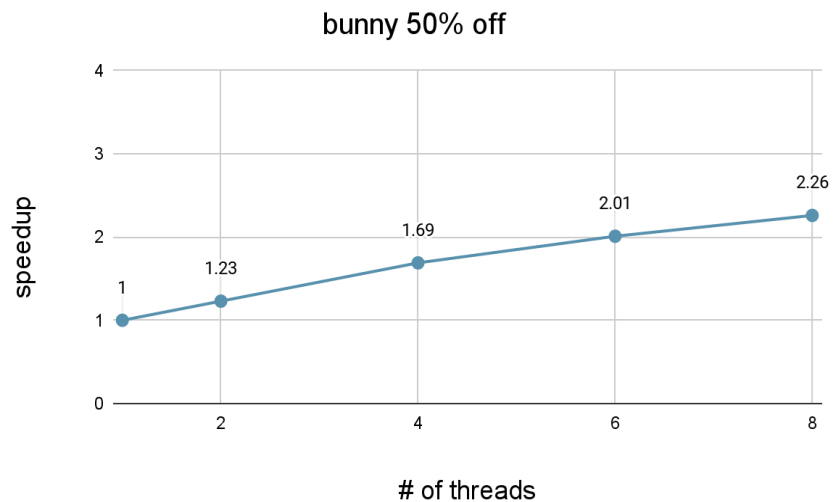


Methodology - OpenMP

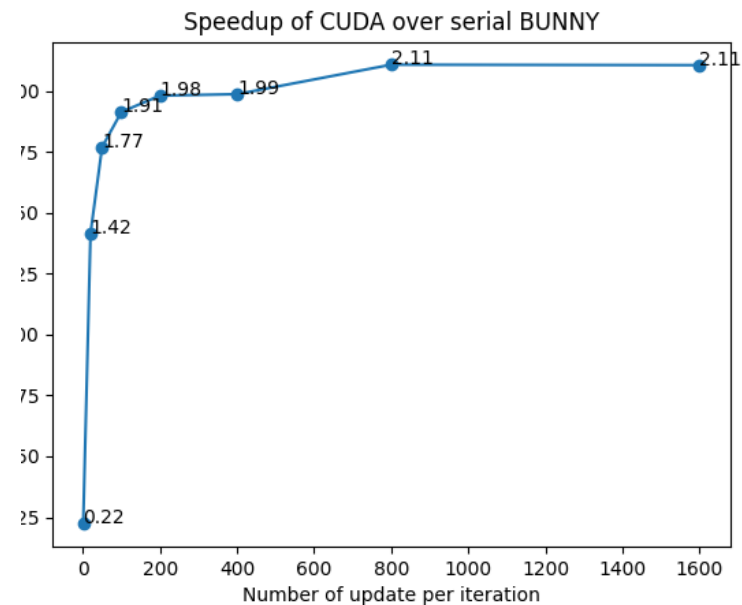
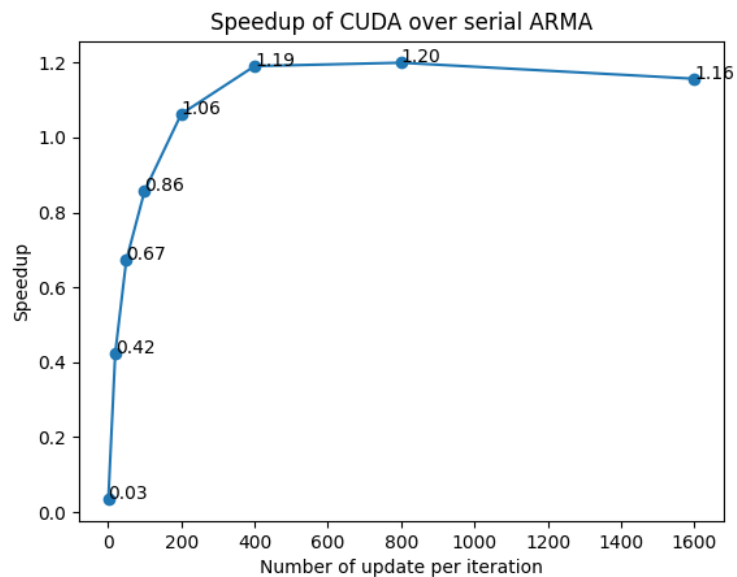
- Divide the mesh into smaller blocks
- Merge vertex pairs in parallel across blocks
- Challenge:
 - Ensure consistency in global simplification



Evaluation - Pthreads



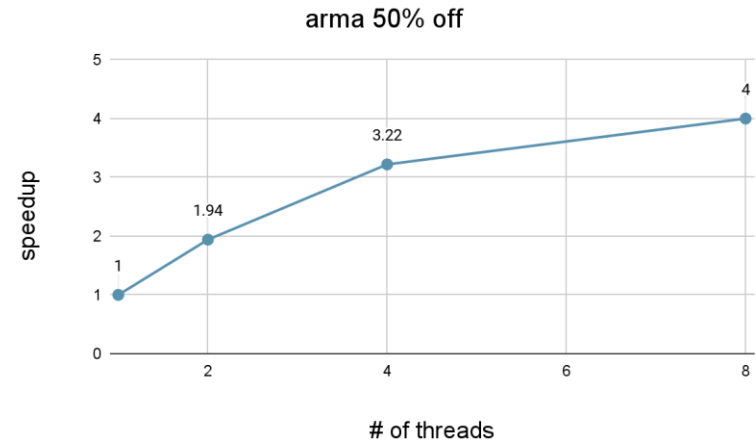
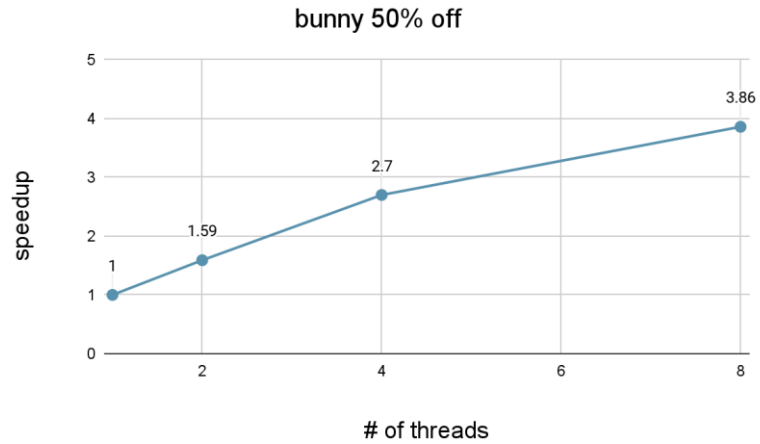
Evaluation - Cuda



Analysis

Total (MB)	Count	Avg (MB)	Med (MB)	Min (MB)	Max (MB)	StdDev (MB)	Operation
140.360	68	2.064	1.067	0.243	18.365	4.121	[CUDA memcpy Device-to-Host]
126.285	68	1.857	0.847	0.121	9.182	2.127	[CUDA memcpy Host-to-Device]

Evaluation - OpenMP



Conclusion & Future work

- Achievements
 - Demonstrated effective parallelization techniques (OpenMP, Pthread, Cuda)
 - Achieved measurable performance improvements
- Challenges
 - Addressed global consistency in simplification.
 - Overhead when loading the data onto the GPU.
- Future work
 - Evaluation of mesh quality.
 - Integration to 3D model software.

Thank you

Q & A